

CREDIT RISK DEFAULT PREDICTION

Executive Summary — Approach, Results & Recommendations

Innovexa Catalyst · Machine Learning Engineering Case Study | Target: **loan_status** (1 = default, 0 = non-default) | Dataset: 32,581 loan applicants, 12 raw features

0.955

Test AUC-ROC

0.825

Test F1-Score (Default)

0.916

PR-AUC (Default class)

21.8%

Base Default Rate

*Final model: **LightGBM**, **Optuna-tuned**, selected over Logistic Regression, Decision Tree, Random Forest, XGBoost and CatBoost baselines.*

1 Approach

The objective was to predict the probability that a loan applicant defaults, using a dataset of 32,581 applicants and 12 raw features covering applicant demographics (age, income, employment length, home ownership), loan characteristics (amount, intent, grade, interest rate) and credit-bureau history. The target is imbalanced — roughly 22% default vs. 78% non-default (about 1:3.6) — which shaped every downstream modeling decision.

1.1 Data Quality & Cleaning

Exploratory analysis surfaced several data-quality issues that were resolved before modeling:

- **Missing values:** `loan_int_rate` (9.6% missing) and `person_emp_length` (2.75% missing) — addressed with median imputation, fit on the training split only to avoid leakage.
- **Duplicates:** 165 exact duplicate rows (0.51% of data) removed.
- **Implausible values:** ages up to 144 years and employment lengths up to 123 years were identified as data-entry errors; 5 rows with age > 100 and 2 rows with employment length exceeding a plausible working lifetime were removed. Final clean dataset: 32,409 rows.

1.2 Feature Engineering

Eight engineered features were added to capture affordability and stability signals not explicit in the raw fields, including **`income_to_loan_ratio`** and **`loan_percent_income`** (affordability/leverage), **`credit_history_to_age_ratio`** and **`emp_length_to_age_ratio`** (behavioral stability), **`age_at_first_credit`**, an ordinal encoding of `loan_grade`, a prior-default flag, and an income-bracket grouping. Nominal categoricals (home ownership, loan intent, income bracket) were one-hot encoded; `loan_grade` and prior default history were encoded numerically since they carry order or binary meaning.

1.3 Modeling Strategy

Data was split 80/20 (stratified on the target) into 25,927 training and 6,482 test rows, preserving the ~21.9% default rate in both splits. Two interpretable baselines (Logistic Regression with balanced class weights, and a depth-limited Decision Tree) were trained first to establish a performance floor. Four advanced models followed — Random Forest, XGBoost, LightGBM and CatBoost — using a mix of **class-weighting** and **SMOTE oversampling** (applied to the training fold only, never the test set) to address the 1:3.6 class imbalance. The strongest candidate, LightGBM, was then tuned with **Optuna** (Bayesian hyperparameter search over tree depth, leaf count, learning rate, subsampling and regularization) to arrive at the final model.

1.4 Evaluation Framework

Because accuracy is misleading under class imbalance, models were compared on AUC-ROC, F1-score, precision/recall for the default class, PR-AUC and the Kolmogorov–Smirnov (KS) statistic. The final model was further validated with 5-fold stratified cross-validation, performance breakdowns by loan-grade and home-ownership segment, and SHAP-based interpretation for both global feature importance and individual applicant explanations.

2 Results

2.1 Model Comparison

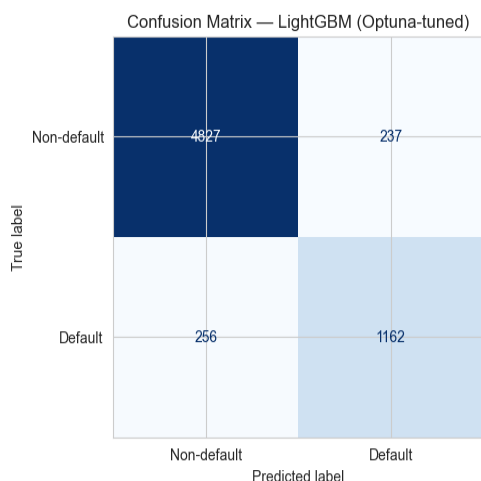
Seven models were evaluated on the held-out test set (6,482 applicants). The Optuna-tuned LightGBM model achieved the best performance across every metric:

Model	AUC-ROC	F1-Score	Precision	Recall	PR-AUC	KS
LightGBM (Optuna-tuned)	0.955	0.825	0.831	0.819	0.916	0.775
LightGBM (weighted)	0.953	0.823	0.830	0.817	0.911	0.773
CatBoost (weighted)	0.944	0.802	0.803	0.802	0.899	0.748
XGBoost (SMOTE)	0.938	0.818	0.920	0.737	0.891	0.740
Random Forest	0.930	0.792	0.823	0.764	0.880	0.721
Decision Tree	0.906	0.778	0.807	0.750	0.857	0.704
Logistic Regression	0.869	0.629	0.524	0.786	0.685	0.592

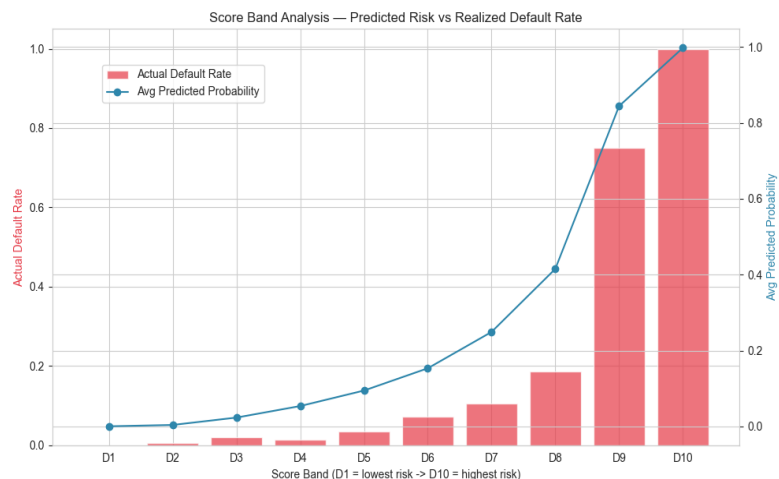
Metrics for the default (minority) class. The tuned LightGBM model was validated with 5-fold stratified cross-validation: AUC-ROC 0.948 ± 0.002 , F1 0.811 ± 0.008 , confirming stable, non-overfit performance.

2.2 Classification Performance

On the test set, the final model correctly separates risk well in both directions: it identifies 82% of true defaulters (recall) while 83% of applicants it flags as default actually default (precision), giving a balanced F1 of 0.825 — without sacrificing performance on the majority non-default class (95% precision and recall).



Confusion matrix — tuned LightGBM, test set (n=6,482)

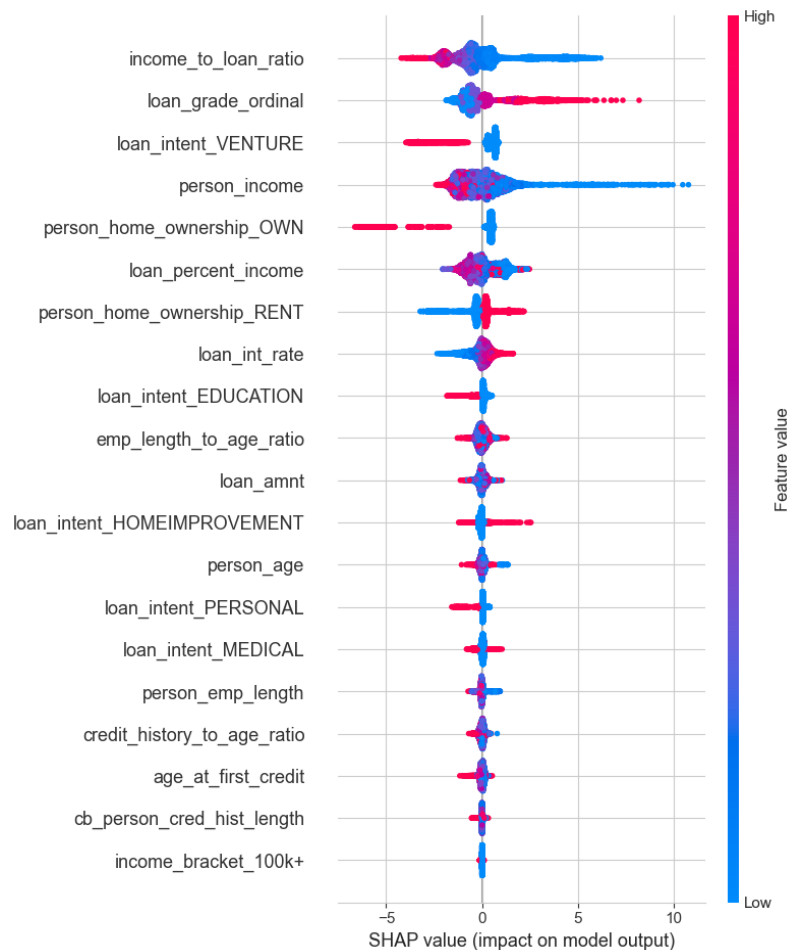


Score-band analysis: predicted risk deciles vs. realized default rate

The decile chart confirms the model is well rank-ordered: the bottom 50% of applicants by score (bands D1–D5) default at under 4%, while the riskiest decile (D10) defaults at 100% and the next riskiest (D9) at 75% — a clean monotonic rise that makes the score usable directly for risk-tiered decisioning.

2.3 What Drives Default Risk (SHAP)

SHAP analysis on the test set ranks **income_to_loan_ratio** as the single strongest predictor, followed by **loan_grade**, **loan_intent = VENTURE**, **person_income**, and **home ownership status**. Low income-to-loan ratio, lower (riskier) loan grades, renting rather than owning, and higher loan_percent_income all push predictions toward default; higher income and home ownership push toward non-default.



SHAP beeswarm — feature impact on predicted default probability (red = high feature value, blue = low feature value)

2.4 Stability Across Customer Segments

The model was stress-tested across loan-grade and home-ownership segments to confirm it generalizes rather than exploiting one dominant pattern:

Loan Grade	N	Default Rate	AUC-ROC
A	2,172	9.7%	0.950
B	2,103	15.8%	0.935
C	1,215	22.6%	0.905
D	729	59.0%	0.977
E	208	61.1%	0.959
F	45	71.1%	0.966

Home Ownership	N	Default Rate	AUC-ROC
RENT	3,275	31.4%	0.967
MORTGAGE	2,651	12.9%	0.899
OWN	533	7.9%	1.000
OTHER	23	21.7%	0.933

AUC-ROC stays at or above 0.90 in nearly every segment — discrimination is strongest in the highest-risk grades (D–F) and weakest, though still strong, in MORTGAGE applicants. Default rates rise sharply for lower grades (A: 9.7% → F: 71.1%) and for renters versus owners (31.4% vs. 7.9%), consistent with the EDA and with the SHAP findings above — giving confidence the model has learned genuine risk signal rather than overfitting to noise.

2.5 Deployment Artifact

The final tuned LightGBM model and its expected feature columns were serialized (final_credit_risk_model_lgbm.pkl) along with a single-applicant scoring function that returns a predicted default probability, a risk tier, and the top SHAP-driven factors behind each individual prediction — making the model usable both for batch portfolio scoring and real-time, explainable underwriting decisions on new applicants.

3 Recommendations

Based on model performance, feature drivers, and segment behavior, the following actions are recommended for credit-risk strategy and next-stage model development.

1 Deploy the tuned LightGBM model as the primary scoring engine

With AUC-ROC 0.955 and stable 5-fold cross-validated performance (0.948 ± 0.002), the model materially outperforms a logistic-regression baseline (AUC-ROC 0.869) while remaining explainable via SHAP. It is ready for use as the core PD (probability of default) model, with the saved .pkl artifact and scoring function enabling integration into underwriting workflows.

2 Adopt risk-tiered decisioning using the score-band structure

The decile analysis shows near-zero realized default in bands D1–D5 (<4%) versus 75–100% in D9–D10. Rather than a single approve/decline cutoff, use these bands to set tiered pricing or terms: auto-approve at standard rates in D1–D5, manual review or risk-based pricing in D6–D8, and decline or require collateral/co-signers in D9–D10.

3 Prioritize income-to-loan ratio and loan grade in underwriting policy

These are the two strongest SHAP drivers of predicted default. Underwriting guidelines and pricing should continue to weight affordability (loan amount relative to income) and bureau-assigned grade heavily, and loan officers should be trained to flag applications with low income-to-loan ratios for closer review even when other factors look favorable.

4 Apply extra scrutiny to renters and VENTURE-purpose loans

Renters default at 31.4% versus 7.9% for homeowners, and VENTURE-purpose loans were a top SHAP driver of risk. This doesn't justify blanket exclusion, but supports adding stipulations (smaller loan sizes, higher rates, or added documentation) for these segments, particularly when combined with a low income-to-loan ratio.

5 Fix data-collection issues at the source

9.6% of interest-rate values and 2.75% of employment-length values were missing, and implausible ages (up to 144) and employment lengths (up to 123 years) had to be removed. Tightening input validation at loan origination will reduce reliance on imputation and improve both data quality and model accuracy over time.

6 Monitor and extend the model post-deployment

Because MORTGAGE applicants showed the relatively lowest (though still strong) AUC-ROC (0.899), this segment is the best candidate for first-line monitoring as new data arrives. Recommended next steps: set up periodic recalibration as repayment outcomes mature, track population stability (PSI) on top SHAP features, and evaluate adding external bureau attributes or macroeconomic indicators to improve coverage of edge-case applicants.

Source: `credit_risk_default_prediction_final.ipynb` — Innovexa Catalyst credit risk case study notebook (dataset: `credit_risk_dataset.csv`, 32,581 applicants).